

Topic:

Einstein Postulates of special theory of relativity

In 1905, Albert Einstein created the theory of special relativity based on two main rules, which are called postulates. Here are the notes on how it works in simple language:

1. The Two Main Rules (Postulates)

The First Rule (Principle of Relativity): This says that the laws of physics are the same for everyone, as long as they are moving at a steady speed in a straight line. Whether you are standing still or riding on a fast, steady train, the way science works does not change.

The Second Rule (Speed of Light): This says that light always travels at the same speed (c) in empty space. It does not matter how fast the person who turned on the light is moving—the light speed stays exactly the same for everyone.

2. Hidden Assumptions

Even though Einstein only listed two main rules, he also had to assume a few other things were true to make his math work:

Space is the same everywhere: The rules of the universe don't change just because you moved to a different spot.

Space is the same in every direction: It doesn't matter which way you are facing; physics works the same way.

3. The Idea of "Spacetime"

Einstein's theory treats space and time as one single thing called "spacetime".

In this theory, every point is an "event".

The path an object takes through this 4D world is called its "worldline".

Even if two people see distances or time differently, they will both agree on a special measurement called the "spacetime interval".

4. Is it Enough?

Some scientists argue that Einstein's two rules are not enough to explain everything.

Empty Space vs. Real Life: The rules were mostly written for empty space (a vacuum).

The "Bridge" Problem: Some believe you need an extra "bridging rule" to explain how light works when there is "stuff" (matter) in the way, like air or glass.

5. Old Physics vs. New Physics

Galilean Relativity (Old): Before Einstein, people used the ideas of Isaac Newton. This works fine for slow things, but it fails when things move close to the speed of light.

Special Relativity (New): This is the more accurate version of physics. When things move very slowly, Einstein's rules start to look just like the old rules. For example, the famous equation $E=mc^2$ comes from combining these relativity rules with old-fashioned science.